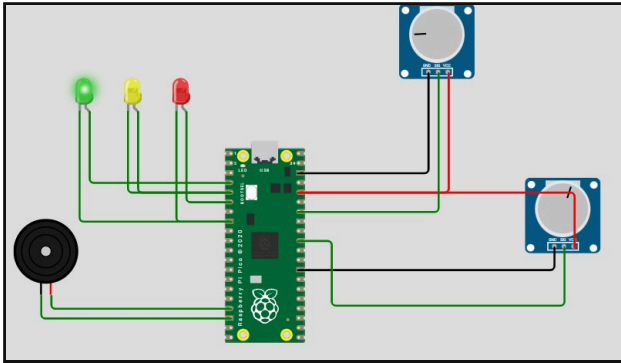
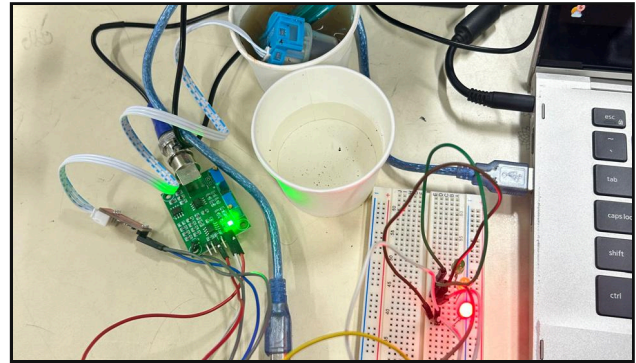


SMART WATER QUALITY MONITORING SYSTEM



ONLINE SIMULATION MODEL



HARDWARE WORKING MODEL

ABSTRACT

This project presents a Smart Water Quality Monitoring System that checks the quality of water using pH and turbidity sensing. The system reads the sensor values through a microcontroller and classifies the water condition as safe, warning, or unsafe. Based on the measured values, green, yellow, and red LEDs are used for visual indication, and a buzzer is activated during unsafe conditions. The project was first tested using an online simulation and then implemented using hardware components. This system provides a simple, low-cost, and useful method for continuous water quality observation in laboratories, homes, and small-scale water monitoring applications.

COMPONENTS REQUIRED

- Arduino UNO / Raspberry Pi Pico
- pH Sensor Module
- Turbidity Sensor Module
- Green, Yellow and Red LEDs
- Buzzer Module
- Breadboard
- Jumper Wires
- USB Cable and Water Sample

SOFTWARE REQUIREMENTS

- Arduino IDE
- Wokwi Online Simulator
- Embedded C / MicroPython
- Serial Monitor for output verification

WATER QUALITY CONDITIONS

Parameter	Safe	Warning	Unsafe
pH Value	6.5 - 8.5	5.0 - 6.4 / 8.6 - 9.0	< 5.0 or > 9.0
Turbidity	Low	Medium	High

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SYSTEM DIAGRAM

